Customer Churn Prediction Report

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Project Description

In the competitive landscape of e-commerce, customer retention is a critical driver of profitability. Acquiring new customers is significantly more expensive than retaining existing ones, and high churn rates can severely impact revenue and long-term business sustainability. As such, understanding and predicting customer churn has become a strategic priority for online retailers.

This project focuses on analyzing customer churn for a leading e-commerce company, using a publicly available dataset from Kaggle. The dataset includes 5,630 observations and 20 features representing customer demographics, activity patterns, and behavioral characteristics. The aim is to build predictive models that can identify customers at risk of churning and uncover the key factors contributing to customer attrition.

The analysis is designed to answer the following business questions:

1. **Which customers are most likely to churn?**
   By identifying patterns and characteristics of customers who have churned, the models will highlight which customers are at risk of leaving.
2. **What factors contribute to customer churn?**
   The analysis will examine a variety of features, such as order frequency, customer satisfaction scores, device type, and usage behavior, to determine which variables are most strongly associated with churn.
3. **How does customer behavior and satisfaction correlate with retention?**
   A deeper look into engagement metrics, including app usage, order count, and customer support interactions, will help the business understand how customer satisfaction and behavior influence loyalty. These findings can inform efforts to improve the overall customer experience.

To achieve this, we developed and compared three machine learning models: K-Nearest Neighbors (KNN) and Random Forest (both implemented in R), and a classification model using SAP Analytics Cloud. These models allowed for a comprehensive approach to predictive analytics using both programming-based and business intelligence platforms. Each model was used to answer all three business questions. Using each model to explore the same questions allowed us to compare their predictive performance and assess how interpretable and actionable the results were across different techniques.

Through this analysis, we aim to uncover actionable insights that will help the business reduce churn rates, enhance customer satisfaction, and ultimately increase customer lifetime value by enabling informed and proactive retention strategies.

Data Preprocessing

1. **Data cleaning:**

We identified and handled missing data as part of the cleaning process. While there were no errors in the data, we encountered missing values in several columns. Specifically, we found blank rows in the following variables: *Tenure, HoursSpendOnApp, OrderAmountHikeFromLastYear, CouponsUsed, OrderCount, and DaySinceLastOrde*r.
To address this, we deleted the rows with missing values in these columns. This reduction left us with a dataset of 4,025 rows out of the original 5,630, meaning we removed approximately 1,605 rows.

2. **Data extraction:**

In terms of data extraction, we identified irrelevant variables that were unlikely to contribute to our analysis of customer churn and removed them. These variables include: *CityTier, PreferredLoginDevice, NumberofDevicesRegistered, and WarehouseToHome*.
Moreover, as part of the data cleaning process documented above, we removed the previously mentioned rows because they contained blanks.

3. **Data transformations:**

For data transformation, we focused on converting categorical variables into dummy variables to make them usable in a machine learning workflow:

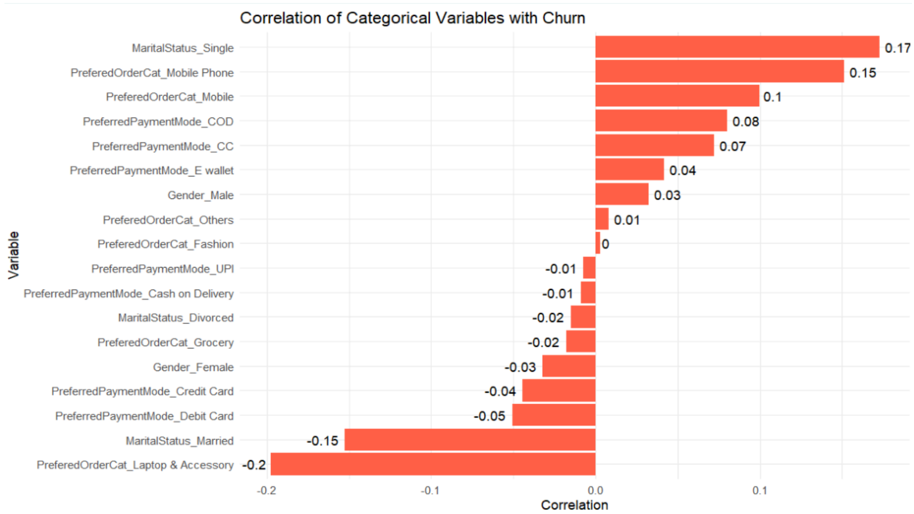
1. **PreferredPaymentMethod** was grouped into two dummy variables:
    - *Mode_Cash* (representing “COD” and “Cash On Delivery”)
    - *Mode_E-payment* (representing “Debit Card”, “Credit Card”, “E-wallet”, and “UPI”)
2. **Gender** was transformed into two binary dummy variables:
    - *Is_Male*
    - *Is_Female*
3. **MaritalStatus** was converted into three dummy variables:
    - *Is_Single*
    - *Is_Married*
    - *Is_Divorced*
4. **PreferredOrderCat** was split into six dummy variables to represent various product categories:
    - *Cat_Fashion*
    - *Cat_Grocery*
    - *Cat_Laptop*
    - *Cat_Mobile*
    - *Cat_MobilePhone*
    - *Cat_Others*

Currently, for each categorical variable, we have created one dummy variable for each category, except for the payment method, where we grouped similar methods together. This allows us to easily perform a validation check to ensure the transformation is correct. During the machine learning model training, we will exclude one dummy variable from each category to avoid multicollinearity and ensure no redundancy among the dummy variables.

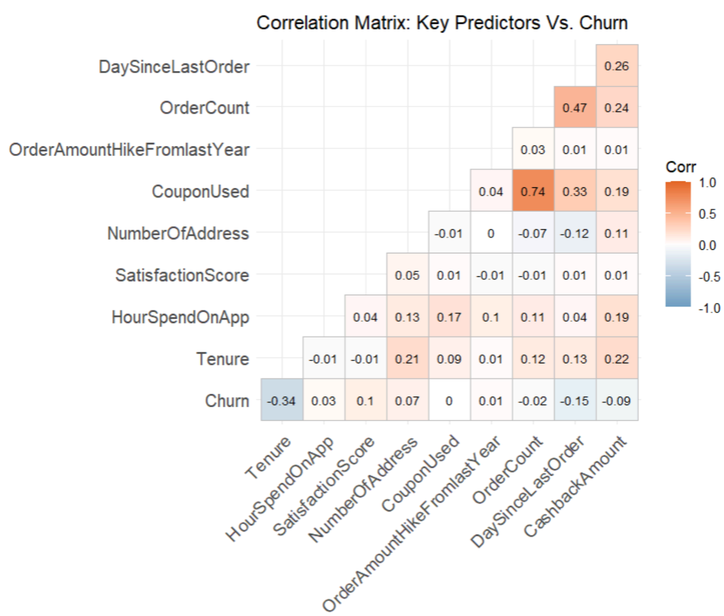
4. **Visualization**

We create 2 correlation charts to see which numerical/categorical variables are most influential to churn.

With categorical variables, people who are single are more likely to churn (+0.17) while people who are married are more likely to stay (-0.15). Moreover, payment methods also impact customers' churn since customers who use COD, CC and E-wallet have a more positive correlation with churn than people who proceed their order through credit or debit card.



For numerical variables, we see that tenure has the most impact on churn with -0.34. This indicates that the longer the customer uses the service, the less likely they are to churn. The next variable that might have a lower impact on churn is Day since last order, with -0.15, indicating that recent purchases are negatively associated with churn - recency of activity is a relevant behavioral signal.

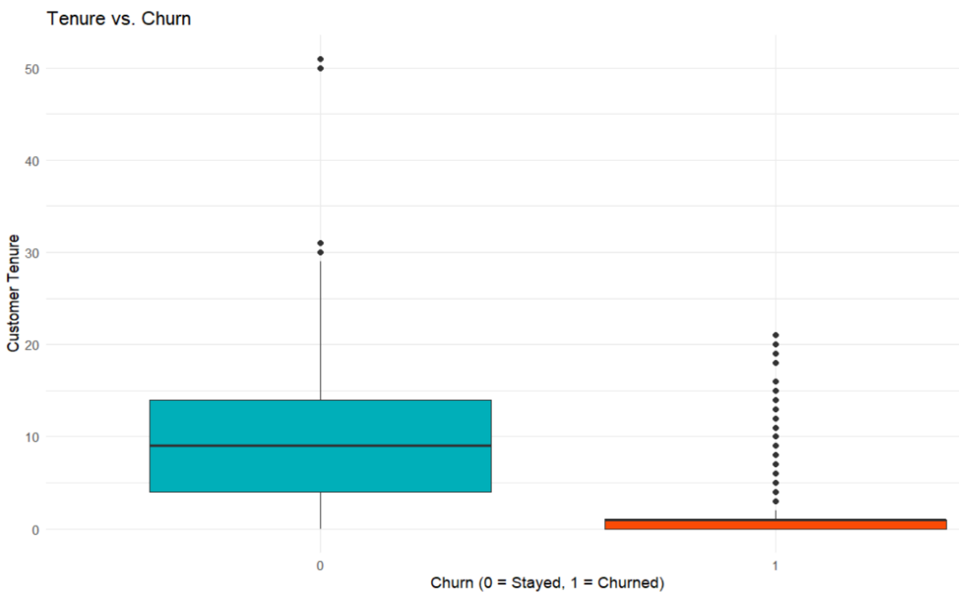


The Tenure vs. Churn boxplot proves that what we found above is true:

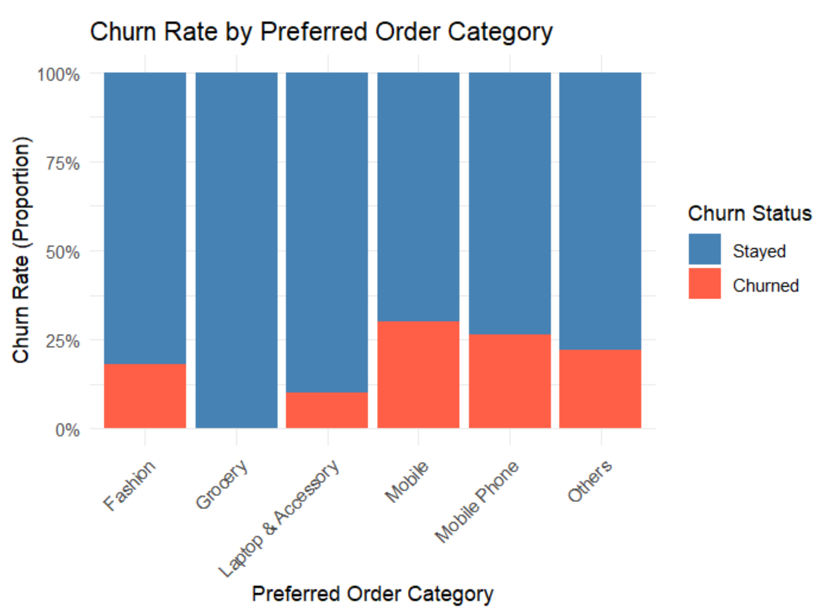
- Customers who churned (Churn = 1) had very short tenure, with most having been with the company for only a very short time

- In contrast, retained customers (Churn = 0) show a wider spread of tenure values, with a higher median and more long-term engagement

This suggests that new customers are significantly more likely to churn, indicating that early lifecycle engagement may be key to improving retention.



Customers preferring Mobile and Mobile Phone categories have the highest churn rates, indicating potential dissatisfaction, high competition, or transient buying behavior. In contrast, those in the Laptop & Accessory category, which is surprising, and Grocery categories exhibit the lowest churn rates, suggesting higher customer retention and possibly stronger brand engagement (with certain technology brands) or recurring purchasing habits (weekly/monthly grocery shopping). We suggest the company would also target mobile/mobile phone customers since they are also in the technology domain.



Models and Analyses

I. K-Nearest Neighbors (KNN) Model:

1. Description of the Technique and Justification

K-Nearest Neighbors (KNN) is a non-parametric, instance-based machine learning algorithm commonly used for classification tasks. It works on the principle that similar data points are likely to be close to each other in the feature space. In the case of churn prediction, KNN identifies the ‘k’ nearest neighbors of a given customer based on a chosen distance metric (we used Euclidean distance in this project) and classifies the customer based on the majority class of these neighbors.

We selected K-Nearest Neighbors (KNN) for its simplicity and effectiveness as a non-parametric machine learning method, as it does not assume any specific distribution of the data. KNN is particularly well-suited for classification tasks such as predicting customer churn. Unlike other models that generate complex decision boundaries or rely on probabilistic assumptions, KNN classifies a data point based on the majority class of its nearest neighbors. This allows us to capture local patterns and similarities between customers. By analyzing the behavior of nearby (similar) customers, KNN helps identify patterns that may influence a customer’s decision to stay or leave that offers intuitive insights into customer behavior.

Because KNN relies on distance calculations, feature scaling is essential. We applied min-max normalization to all numeric predictors to ensure equal weighting in the distance metric to allow the model to fairly evaluate similarity between records across different units and ranges.

2. Predictors and Feature Selection:

The output variable is Churn, which indicates whether a customer has left (churned) or stayed.

The predictors selected include both continuous and categorical features, many of which were transformed into dummy variables to suit the algorithm.

The predictors used in the K-Nearest Neighbors (KNN) model include a mix of demographic, behavioral, and transactional features. Tenure indicates how long a customer has been with the company, while Mode_Cash and Mode_E.payment capture the customer’s preferred payment methods. Gender is captured through Is_Male and Is_Female, while HourSpendOnApp reflects customer engagement with the platform. We also included product category preferences such as Cat_Fashion, Cat_Grocery, Cat_Laptop, Cat_Mobile, Cat_Mobile.Phone, and Cat_Others, which provide insights into customer purchasing behavior. SatisfactionScore measures overall customer satisfaction, and Complain tracks whether a customer has filed any complaints. Additional predictors include marital status (Is_Single, Is_Married, Is_Divorced), NumberOfAddress, and OrderAmountHikeFromLastYear, reflecting any increase in spending. Finally, CouponUsed, OrderCount, DaySinceLastOrder, and CashbackAmount offer insights into promotional engagement, activity frequency, and financial incentives that may influence churn behavior.

We selected these variables based on their potential relationship to churn rate as identified at the beginning of the project. The dataset included a wide range of features, and we kept a comprehensive set of predictors to ensure the model captures all relevant patterns.

II. Random Forest Model:

1. Description of the Technique and Justification

Random Forest is a common supervised machine learning algorithm used for both classification (predicting categorical variables) and prediction (predicting numerical variables). As an ensemble learning method, the model builds a collection of decision trees, each trained on different random subsets of the data-and combines their outputs to produce a more accurate model than an individual tree. In this analysis of predicting customer churn, Random Forest can help determine how accurately we can predict whether a customer will discontinue shopping with the online E-commerce company based on various factors.

The model process starts with multiple random samples of the data being drawn with replacement from the original dataset, which means that some records may appear multiple times in a sample. This helps to create diverse trees that are less likely to make the same errors and improve the model's overall accuracy and stability. After the random samples are drawn, the model will train each decision tree on each random subset of the training data, this process is also known as "Bootstrap". Finally, the classification from each tree will be combined to produce the model's final prediction, which is the majority prediction from a large number of decision trees, and this is also why the model is named Random Forest.

2. Predictors and Feature Selection:

The outcome variable is Churn, which indicates whether a customer has left (churned) or stayed. The selected 19 predictors only include continuous features and dummy variables that were transformed from the categorical variables. Since this is a binary classification, all the categorical columns were removed from the dataset, which are PreferredPaymentMode, Gender, PreferredOrderCat, and MaritalStatus. In addition, one column from each set of dummy variables that were created when converting categorical variables is also dropped, which are Mode_E-payment, Is_Female, Cat_Others, and Is_Divorced. By dropping these columns, multicollinearity can be avoided and ensure the stability of the model's performance.

Before running the model, data preprocessing is needed to turn the outcome variable, Churn, from a numerical to a categorical variable with 0 indicating that the customer has stayed, while 1 means the customer has left. Since the goal is classifying the data, leaving the outcome variable as numerical will produce incorrect results, which will predict numbers rather than classification; therefore, the outcome variable is transformed to a categorical variable instead.

III. SAP Classification Model

1. Description of the Technique and Justification

SAP Classification Model based on Gradient Boosting that involves building a strong model from a collection of weaker models. As a form of ensemble learning, SAP classification model takes advantage of strength in numbers as it leverages multiple weak decision trees to produce a strong prediction of the data. Gradient Boosting builds trees sequentially with each new tree focusing on correcting the errors of the previous ones by minimizing a loss function. This iterative process results in a strong predictive model.

SAP classification models iteratively compute the difference between the current prediction and the actual target values, then train a weak learner on the residuals and add new trees to improve the performance of the model. After updating the model's prediction by combining the previous predictions with new models, it then repeats the iterating process until reaching a maximum number of iterations or predefined level of accuracy. In predicting the churn likelihood, SAP Classification model is suitable since it often achieves high accuracy, can handle various types of loss functions, is able to learn from previous models' errors, and robust to outliers and missing data.

2. Predictors and Feature Selection:

The primary objective of the model is to predict the likelihood of customer churn—a binary outcome represented by the variable Churn, where 1 indicates that a customer has left the service, and 0 indicates the customer continues with the service. Before training, the Churn variable was converted from a numeric format to a categorical variable to ensure that the model treated it appropriately as a classification target rather than a continuous value.

The final model used a total of 19 predictors, selected based on domain relevance, data quality, and the need to reduce redundancy and multicollinearity. These predictors include a combination of continuous variables, such as Tenure, HourSpendOnApp, CashbackAmount, and OrderAmountHikeFromLastYear, as well as binary dummy variables derived from original categorical features.

To handle the categorical data effectively, features like PreferredPaymentMode, Gender, PreferredOrderCat, and MaritalStatus were transformed into dummy variables. However, to prevent multicollinearity, one dummy variable from each categorical group was excluded, such as Mode_E-payment, Is_Female, Cat_Others, and Is_Divorced.

Results and discussion

I. K-Nearest Neighbors (KNN) Model:

To prepare the dataset for training and evaluation, we partitioned the original dataset (consisting of 4,025 records) into training, validation, and test sets using random sampling via the sample() function in R. We allocated 80% of the data (3,220 records) to the training set, 10% (402 records) to the validation set, and the remaining 10% (403 records) to the test set. After training the KNN model on the training data, we found that k = 5 yielded the best results, with an overall accuracy of 85.34% on the training set.

After training the model on the training data, the model's performance was evaluated on both the validation and test datasets that yield the following key metrics. On the validation set, the model achieved an accuracy of 86.32%, with a sensitivity of 42.86% and a specificity of 95.48%. Sensitivity measures the model's ability to correctly identify churners, meaning that only 42.86% of the customers who actually churned were correctly predicted. Specificity, on the other hand, measures the model's ability to correctly identify non-churners, and the model performed well in this regard with a specificity of 95.48%. This indicates that the model correctly classified 95.48% of the customers who did not churn. The precision was 66.67%, meaning that among the customers predicted to churn, approximately 66.67% actually did.

On the test set, accuracy was essentially the same, slightly increasing to 86.82%, while sensitivity increased marginally to 44.29%, and specificity remained very similar at 95.78%. Precision also increased to 68.89%, indicating a better proportion of true churn predictions. Overall, while the model excels in minimizing false positives (non-churn customers predicted as churn), there is room to improve the detection of actual churn cases. Since detecting churn is the primary objective, the model's current performance may not be optimal.

To better understand which predictors contribute most to the model's predictions, we used the varImp() function to assess the importance of each variable. Based on the analysis, the key factors driving customer churn include Tenure, Complain, DaySinceLastOrder, Cat_Laptop, and Is_Single. Customers with shorter Tenure are more likely to churn, while long-term customers tend to stay. Complaints and long gaps between orders (DaySinceLastOrder) were also identified as strong predictors, suggesting that dissatisfied customers or those who have not engaged recently are at a higher risk of leaving. Interestingly, customers who frequently purchase laptops (Cat_Laptop) exhibited distinct churn behaviors, which could indicate that this group has specific preferences or needs. Moreover, the significance of Is_Single suggests that personal circumstances may influence retention, with single customers potentially showing different churn patterns compared to others.

II. Random Forest Model:

Similar to the K-Nearest Neighbors model, 80% of the data (3,220 records) goes into the training dataset, while the remaining 20% (805 records) to validation and test from the original dataset consists of 4,025 records. Among the 20%, 10% to the validation set, which is equivalent to 402 records, and the remaining 10% consists of 403 records to the test set.

By using the Random Forest algorithm, Ranger, sample data of 3,220 records were randomly drawn with replacement from the training data. Each sample was used to construct its decision tree, resulting in a total of 3,220 decision trees in the model. After training our Random Forest model, the prediction performance on the validation and test sets are ready to be evaluated. On the validation set, the overall accuracy of 98.51% indicates that the model correctly predicted customer behavior 98.51% of the time, demonstrating high accuracy in distinguishing between customers who stayed and those who churned. In terms of the sensitivity, the model successfully identified 93.55% of actual churned customers who are likely to leave from the dataset. For customers who stayed, the specificity of 99.41% shows that the model correctly identifies loyal customers. Another important metric in the model's performance is the precision, which tells the level of confidence in the model's churn predictions. Among all customers who are predicted to churn, 96.67% actually did. On the test set, the accuracy is slightly lower at 97.52% while the sensitivity is higher at 94.87%. Compared to the model's performance on the validation set, specificity in the test set is decreased to 98.15% as well as the precision also decreases by approximately 4% to 92.50%. From the test set key metrics, it is observable that only the sensitivity is better, which indicates that the model predicts actual churned customers who are likely to leave from the dataset more accurately. However, the other three metrics, which are accuracy, specificity, and precision, are slightly lower. Although the models' key metrics are slightly different on the test set, the model performance is still strong.

Overall, the Random Forest model demonstrates outstanding performance across all key metrics. Compared to the K-Nearest Neighbors model, Random Forest performs better in predicting customer churn by having a significantly higher percentage in those key metrics, making it a more effective model for this task.

III. SAP Classification Model:

To prepare the dataset for training and test set, we manipulated the original dataset in Excel using a random function. Using the random function, about 80% of the data (3,213 records) goes into the training dataset, while the remaining 20% (812 records) to validation and test from the original dataset consists of 4,025 records.

On the validation set, the model achieved an accuracy of 92%, indicating that it was able to correctly predict customer churn behavior in a substantial majority of cases. Sensitivity on this set was 79%, meaning the model identified 79% of the customers who actually churned, while the specificity was 95%, showing strong performance in identifying customers who stayed. Precision stood at 78%, and the F1 score was 0.79, indicating that the model is decent.

When applied to the test data, the SAP model demonstrated improved performance across all key metrics. The accuracy increased to 97.17%, suggesting a very high rate of correct predictions overall. Sensitivity, or true positive rate, improved to 89.80%, showing a strong capability in detecting churners. Specificity reached 98.79%, further confirming the model's precision in correctly identifying non-churners. In terms of precision, the model scored 94.29%, meaning that nearly all customers predicted to churn did in fact churn. The F1 score also rose to 0.92, indicating a well-balanced and effective classification model.

Compared to Random Forest and kNN models, SAP Classification Model performs pretty similar to Random Forest with high accuracy and specificity. The precision of the SAP model is slightly higher than Random Forest with 94.29% than 92.5%. However, the sensitivity of Random Forest is higher at 94.87% compared to 89.80% of the SAP model.

IV. Recommended actions:

Based on the findings from the three models, the business should consider implementing targeted retention strategies to reduce churn. To address churn risk for customers with shorter Tenure, the business could implement early engagement strategies, such as personalized onboarding or loyalty programs to increase customer attachment. For customers who have expressed complaints, proactive customer service interventions like quick resolutions or satisfaction surveys could help reduce dissatisfaction and the likelihood of churn. Re-engaging customers who have not ordered recently (DaySinceLastOrder) with personalized offers or reminders could bring them back into the fold. For the specific segment of customers who frequently purchase laptops, tailored promotions or exclusive offers for related products may help maintain their interest. Finally, the importance of Is_Single suggests that single customers may have unique churn behaviors, so creating personalized campaigns or offering relationship-building incentives (e.g., referral bonuses or exclusive member benefits) for this group could be beneficial in reducing their churn risk.

Summary

By applying multiple machine learning models to assess the ecommerce customer churn dataset, our group identified key factors influencing churn and evaluated which models best support different business objectives. The choice of model depends on the company's priorities: if the goal is to accurately predict which customers are likely to churn, the Random Forest model offers higher performance on accuracy, sensitivity, and precision metrics. For businesses interested in exploring and understanding patterns behind customer behavior, the K-Nearest Neighbors (KNN) model provides valuable insights. Similar to Random Forest, SAP Classification model can also be used to predict the likelihood of customer churn.

Our analysis highlighted factors that are strongly correlated with customer churn, including Tenure, Complaints, DaySinceLastOrder, Cat_Laptop, and Marital Status (Is_Single). Specifically, long-term customers tend to stay, while those with shorter tenure are more likely to leave. In addition, customer satisfaction also correlated with retention rate, as dissatisfied customers who raised complaints recently show a higher risk of churn. As the duration since the last orders gets longer, customer engagement also declines. Besides these expected variables that affect customer churn rate, customers who are single and frequently purchase laptops also exhibit distinct churn patterns, which suggests that demographic and product preferences also play a role in retention.

By understanding the influences of these major factors on customer churn, e-commerce companies can tailor their retention strategies effectively. For instance, the company can potentially implement loyalty programs to increase customer attachment for those with shorter tenure. The company can address customer complaints through satisfaction surveys and actively take their feedback into consideration if it's feasible. For customers who are single and frequently purchase laptops, the company can create exclusive offerings related to their interests, and relationship-building initiatives may strengthen their loyalty. Implementing specific strategies to address the main factors that contribute to customer retention will help companies better meet customer needs and improve retention rates.

In terms of predictive performance, the Random Forest model outperforms the K-Nearest Neighbors (KNN), making it a stronger choice for forecasting churn for online e-commerce companies. Its higher accuracy and precision help predict at-risk customers and allow companies to intervene through effective strategies. Between SAP Classification and Random Forest models, the sensitivity in Random Forest has a higher percentage while the precision in SAP Classification performs better, however, the performance keys metrics in these two models are very similar with high accuracy and specificity. This indicates that SAP Classification model is better at predicting consumers that were anticipated to churn actually did. Meanwhile, Random Forest has higher percentages of accurately forecasted clients who actually churned.

Overall, our analysis demonstrates how machine learning models and data-driven insights can help businesses better understand their customers, identify areas for improvement, and develop strategic growth plans. Data tells a story, and when properly analyzed and applied, it can enhance customer satisfaction and drive business success.